

Groups, Rings, Fields

Math Foundations — Node 04

First, an example

Before the axioms, here is a tiny group you can compute by hand: the integers modulo 4 under addition, $\mathbb{Z}/4\mathbb{Z} = \{0, 1, 2, 3\}$, with $a + b$ taken mod 4. The full Cayley table is

$+$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Read aloud: the table whose rows and columns are indexed by zero, one, two, three, and whose entry in row a column b is a plus b taken modulo four.

Notice three things: 0 acts as identity (the row labelled 0 reproduces the header); every element has an additive inverse on this row (e.g. $1 + 3 = 0$); and the table is symmetric across the diagonal, so $+$ is commutative. These three observations are exactly the group axioms we are about to formalise.

1 Motivation

Abstract algebra studies sets equipped with operations. By isolating the *axioms* that an operation satisfies, we can simultaneously reason about integers, rotations of a cube, residues modulo a prime, invertible matrices, and continuous symmetries of a manifold. Three structures form the spine of the subject: groups, rings, and fields, with each obtained by adding more structure to the previous one.

2 Groups

Definition 1 (Group). A *group* is a pair $(G, \cdot)$ where G is a set and $\cdot : G \times G \rightarrow G$ is a binary operation satisfying:

1. **Associativity:** $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in G$.
2. **Identity:** there exists $e \in G$ with $e \cdot a = a \cdot e = a$ for every $a \in G$.
3. **Inverses:** for every $a \in G$ there exists $a^{-1} \in G$ with $a \cdot a^{-1} = a^{-1} \cdot a = e$.

A group is *abelian* if additionally $a \cdot b = b \cdot a$ for all $a, b \in G$.

Example 1. $(\mathbb{Z}, +)$ is an abelian group with identity 0 and inverses $n \mapsto -n$. The symmetric group S_n of permutations of n symbols is a non-abelian group for $n \geq 3$.

Theorem 1 (Uniqueness of the identity). *In a group $(G, \cdot)$, the identity element is unique.*

Proof. Suppose e and e' are both identity elements of G . Applying the identity axiom for e' to the element e , we have $e \cdot e' = e$. Applying the identity axiom for e to the element e' , we have $e \cdot e' = e'$. Therefore

$$e = e \cdot e' = e',$$

Read aloud: e equals e dot e -prime, which equals e -prime; that is, the two candidate identities collapse to the same element via one application of each identity axiom.

so $e = e'$, and the identity element is unique. $\square$

Proposition 2 (Uniqueness of inverses). *For each $a \in G$ the inverse a^{-1} is unique.*

Proof. Suppose b and c both satisfy $a \cdot b = b \cdot a = e$ and $a \cdot c = c \cdot a = e$. Then

$$b = b \cdot e = b \cdot (a \cdot c) = (b \cdot a) \cdot c = e \cdot c = c,$$

Read aloud: b equals b dot e , equals b dot the quantity a dot c , equals the quantity b dot a dot c , equals e dot c , equals c ; in words, insert the identity, replace it by $a \cdot c$, reassociate, collapse $b \cdot a$ to e , and strip the identity to conclude $b = c$.

using associativity (in the middle) and the identity axiom (at the ends). $\square$

3 Rings

Definition 2 (Ring). A *ring* is a triple $(R, +, \cdot)$ such that

1. $(R, +)$ is an abelian group, with identity denoted 0.
2. $(R, \cdot)$ is a monoid: $\cdot$ is associative and admits an identity $1 \in R$ such that $1 \cdot a = a \cdot 1 = a$ for all $a \in R$.
3. **Distributivity:** for all $a, b, c \in R$,

$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (b + c) \cdot a = b \cdot a + c \cdot a.$$

Read aloud: a dot the quantity b plus c equals a dot b plus a dot c , and the quantity b plus c dot a equals b dot a plus c dot a ; that is, multiplication distributes over addition on both the left and the right.

A ring is *commutative* if $a \cdot b = b \cdot a$ for all $a, b \in R$.

Example 2. $(\mathbb{Z}, +, \cdot)$ is a commutative ring. The ring $M_n(\mathbb{R})$ of $n \times n$ real matrices, with matrix addition and multiplication, is a non-commutative ring for $n \geq 2$.

4 Fields

Definition 3 (Field). A *field* is a commutative ring $(F, +, \cdot)$ in which $0 \neq 1$ and every nonzero element has a multiplicative inverse: for every $a \in F$ with $a \neq 0$ there exists $a^{-1} \in F$ such that $a \cdot a^{-1} = 1$.

Example 3. $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are fields. The ring $\mathbb{Z}/p\mathbb{Z}$ is a field whenever p is prime; for composite n the ring $\mathbb{Z}/n\mathbb{Z}$ has zero divisors and is not a field.

5 Why Fields Matter for ML

A vector space is, by definition, an abelian group with scalar multiplication by a field. Real-valued arithmetic in gradient descent, the covariance matrix inverse in Gaussian likelihoods, and the advantage normalisation $\hat{A}_i = (r_i - \mu)/\sigma$ in GRPO all silently use that $\mathbb{R}$ is a field — specifically, that $1/\sigma$ exists whenever $\sigma \neq 0$.

Exercises

1. Prove that $(a \cdot b)^{-1} = b^{-1} \cdot a^{-1}$ in any group.
2. Find a nonzero element of $\mathbb{Z}/4\mathbb{Z}$ with no multiplicative inverse, and conclude that $\mathbb{Z}/4\mathbb{Z}$ is not a field.
3. Show that in any ring R , $0 \cdot a = 0$ for every $a \in R$.